

Module4 lab5

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1)

R1

R1#enable

R1#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#interface f0/0

R1(config-if)#ip address 10.0.0.1 255.255.255.0

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#

*Mar 1 00:01:02.943: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:01:03.943: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config)#interface f1/0

R1(config-if)#ip address 10.0.1.1 255.255.255.252

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#e

*Mar 1 00:01:48.451: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:01:49.451: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R1(config)#end

R2

R2#enable

R2#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#interface f0/0

R2(config-if)#ip address 10.0.0.2 255.255.255.0

R2(config-if)#no shutdown

R2(config-if)#exit

R2(config)#

*Mar 1 00:01:57.855: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:01:58.855: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R2(config)#interface f1/0

R2(config-if)#ip address 10.0.2.1 255.255.255.252

R2(config-if)#no shutdown

R2(config-if)#exit

R2(config)#

*Mar 1 00:02:25.279: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:02:26.279: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R2(config)#end

R3

R3#enable

R3#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R3(config)#interface f0/0

R3(config-if)#ip address 10.0.0.3 255.255.255.0

R3(config-if)#no shutdown

R3(config-if)#exit

R3(config)#inte

*Mar 1 00:02:36.619: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:02:37.619: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R3(config)#interface f1/0

R3(config-if)#ip address 10.0.3.1 255.255.255.252

R3(config-if)#no shutdown

R3(config-if)#exit

R3(config)#

*Mar 1 00:03:04.943: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:03:05.943: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R3(config)#interface e2/0

R3(config-if)#ip address 10.0.4.1 255.255.255.252

R3(config-if)#no shutdown

R3(config-if)#exit

R3(config)#end

R4

R4#enable

R4#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R4(config)#interface f0/0

R4(config-if)#ip address 10.0.1.2 255.255.255.252

R4(config-if)#no shutdown

R4(config-if)#exit

R4(config)#int

*Mar 1 00:03:34.659: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:03:35.659: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R4(config)#interface f1/0

R4(config-if)#ip address 192.168.1.1 255.255.255.0

R4(config-if)#no shutdown

R4(config-if)#exit

R4(config)#end

R5

R5#enable

R5#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R5(config)#interface e2/0

R5(config-if)#ip address 192.168.2.1 255.255.255.0

R5(config-if)#no shutdown

R5(config-if)#exit

R5(config)#inter

*Mar 1 00:04:22.915: %LINK-3-UPDOWN: Interface Ethernet2/0, changed state to up

*Mar 1 00:04:23.915: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/0, changed state to up

R5(config)#interface f0/0

R5(config-if)#ip address 10.0.2.2 255.255.255.252

R5(config-if)#no shutdown

R5(config-if)#exit

R5(config)#inter

*Mar 1 00:04:45.899: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:04:46.899: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R5(config)#interface f1/0

R5(config-if)#ip address 10.0.3.2 255.255.255.252

R5(config-if)#no shutdown

R5(config-if)#exit

R5(config)#end

R6

R6#enable

R6#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R6(config)#interface f0/0

R6(config-if)#ip address 10.0.4.2 255.255.255.252

R6(config-if)#no shutdown

R6(config-if)#exit

R6(config)#

*Mar 1 00:05:11.219: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:05:12.223: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R6(config)#interface e2/0

R6(config-if)#ip address 10.0.5.1 255.255.255.252

R6(config-if)#no shutdown

R6(config-if)#exit

R6(config)#

*Mar 1 00:05:43.411: %LINK-3-UPDOWN: Interface Ethernet2/0, changed state to up

*Mar 1 00:05:44.411: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/0, changed state to up

R6(config)#interface f1/0

R6(config-if)#ip address 192.168.3.1 255.255.255.0

R6(config-if)#no shutdown

R6(config-if)#exit

R6(config)#

*Mar 1 00:06:11.539: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:06:12.539: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R6(config)#end

R7

R7#enable

R7#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R7(config)#interface e2/0

R7(config-if)#ip address 10.0.5.2 255.255.255.252

R7(config-if)#no shutdown

R7(config-if)#exit

R7(config)#

*Mar 1 00:06:07.343: %LINK-3-UPDOWN: Interface Ethernet2/0, changed state to up

*Mar 1 00:06:08.343: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/0, changed state to up

R7(config)#interface f0/0

R7(config-if)#ip address 10.0.6.1 255.255.255.252

R7(config-if)#no shutdown

R7(config-if)#exit

R7(config)#

*Mar 1 00:06:39.347: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:06:40.347: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R7(config)#interface f1/0

R7(config-if)#ip address 192.168.4.1 255.255.255.0

R7(config-if)#no shutdown

R7(config-if)#exit

R7(config)#

*Mar 1 00:07:09.447: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:07:10.447: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R7(config)#end

R8

R8#enable

R8#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R8(config)#interface f0/0

R8(config-if)#ip address 10.0.6.2 255.255.255.252

R8(config-if)#no shutdown

R8(config-if)#exit

R8(config)#

*Mar 1 00:07:01.591: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:07:02.591: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R8(config)#interface f1/0

R8(config-if)#ip address 192.168.5.1 255.255.255.0

R8(config-if)#no shutdown

R8(config-if)#exit

R8(config)#

*Mar 1 00:07:29.955: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:07:30.955: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R8(config)#end

Layer2Switch-1

vIOS-L2-01>enable

vIOS-L2-01#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

vIOS-L2-01(config)#interface vlan 1

vIOS-L2-01(config-if)#ip address 10.0.0.10 255.255.255.0

vIOS-L2-01(config-if)#no shutdown

vIOS-L2-01(config-if)#exit

vIOS-L2-01(config)#ip default-gateway 10.0.0.1

vIOS-L2-01(config)#end

PC1

PC1> ip 192.168.1.2 255.255.255.0 192.168.1.1

Checking for duplicate address...

PC1 : 192.168.1.2 255.255.255.0 gateway 192.168.1.1

PC2

PC2> ip 192.168.2.2 255.255.255.0 192.168.2.1

Checking for duplicate address...

PC2 : 192.168.2.2 255.255.255.0 gateway 192.168.2.1

PC3

PC3> ip 192.168.3.2 255.255.255.0 192.168.3.1

Checking for duplicate address...

PC3 : 192.168.3.2 255.255.255.0 gateway 192.168.3.1

PC4

```
PC4> ip 192.168.4.2 255.255.255.0 192.168.4.1
Checking for duplicate address...
PC4 : 192.168.4.2 255.255.255.0 gateway 192.168.4.1
```

PC5

```
PC5> ip 192.168.5.2 255.255.255.0 192.168.5.1
Checking for duplicate address...
PC5 : 192.168.5.2 255.255.255.0 gateway 192.168.5.1
```

2) Настройка RIP v2 — no auto-summary отключает автоматическое суммирование на границах классов, без этого бы получилась каша — 192.168.1.0 была бы объявлена как класс B

R1

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#no auto-summary
R1(config-router)#network 10.0.1.0
R1(config-router)#exit
R1(config)#end
```

R4

```
R4#enable
R4#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#network 10.0.1.0
R4(config-router)#network 192.168.1.0
R4(config-router)#exit
R4(config)#end
```

R2

```
R2#enable
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#no auto-summary
R2(config-router)#network 10.0.2.0
R2(config-router)#exit
R2(config)#end
```

R3

R3#enable

R3#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R3(config)#router rip

R3(config-router)#version 2

R3(config-router)#no auto-summary

R3(config-router)#network 10.0.3.0

R3(config-router)#exit

R3(config)#end

R5

R5#enable

R5#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R5(config)#router rip

R5(config-router)#version 2

R5(config-router)#no auto-summary

R5(config-router)#network 10.0.2.0

R5(config-router)#network 10.0.3.0

R5(config-router)#network 192.168.2.0

R5(config-router)#exit

3) Настройка OSPF

R1

R1#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#router ospf 1

R1(config-router)#network 10.0.0.0 0.0.0.255 area 0

R1(config-router)#exit

R1(config)#end

R2

R2#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#router ospf 1

R2(config-router)#network 10.0.0.0 0.0.0.255 area 0

R2(config-router)#ne

*Mar 1 01:16:40.139: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.1.1 on FastEthernet0/0 from
LOADING to FULL, Loading Done

R2(config-router)#network 10.0.2.0 0.0.0.3 area 2

R2(config-router)#exit

R2(config)#end

R3

R3#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R3(config)#router ospf 1

R3(config-router)#network 10.0.0.0 0.0.0.255 area 0

R3(config-router)#

*Mar 1 01:17:51.731: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.1.1 on FastEthernet0/0 from LOADING to FULL, Loading Done

*Mar 1 01:17:51.735: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.2.1 on FastEthernet0/0 from LOADING to FULL, Loading Done

R3(config-router)#network 10.0.3.0 0.0.0.3 area 2

R3(config-router)#network 10.0.4.0 0.0.0.3 area 1

R3(config-router)#area 1 stub no-summary

R3(config-router)#exit

R3(config)#end

R5

R5#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R5(config)#router ospf 1

R5(config-router)#network 10.0.2.0 0.0.0.3 area 2

R5(config-router)#network 10.0.3.0 0.0.0.3 area 2

R5(config-router)#network 192.168.2.0 0.0.0.255 area 2

R5(config-router)#exit

R5(config)#end

R6

R6#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R6(config)#router ospf 1

R6(config-router)#network 10.0.4.0 0.0.0.3 area 1

R6(config-router)#network 10.0.5.0 0.0.0.3 area 1

R6(config-router)#network 192.168.3.0 0.0.0.255 area 1

R6(config-router)#area 1 stub

R6(config-router)#exit

R6(config)#end

R7

R7#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R7(config)#router ospf 1

R7(config-router)#network 10.0.5.0 0.0.0.3 area 1

R7(config-router)#network 10.0.6.0 0.0.0.3 area 1

R7(config-router)#network 192.168.4.0 0.0.0.255 area 1

R7(config-router)#area 1 stub

R7(config-router)#exit

R7(config)#end

R8

R8#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R8(config)#router ospf 1

R8(config-router)#network 10.0.6.0 0.0.0.3 area 1

R8(config-router)#network 192.168.5.0 0.0.0.255 area 1

R8(config-router)#area 1 stub

R8(config-router)#exit

R8(config)#end

4)

R1

R1#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#router rip

R1(config-router)#redistribute ospf 1 metric 3

R1(config-router)#exit

R1(config)#router ospf 1

R1(config-router)#redistribute rip subnets

R1(config-router)#exit

R1(config)#end

R2

R2#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#router rip

R2(config-router)#redistribute ospf 1 metric 3

R2(config-router)#exit

R2(config)#router ospf 1

R2(config-router)#redistribute rip subnets

R2(config-router)#exit

R3

R3#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R3(config)#router rip

R3(config-router)#redistribute ospf 1 metric 3

R3(config-router)#exit

R3(config)#router ospf 1

R3(config-router)#redistribute rip subnets

R3(config-router)#exit

R3(config)#end

5)

```
PC5> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=59 time=64.115 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=59 time=50.862 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=59 time=57.403 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=59 time=61.438 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=59 time=73.910 ms

PC5> ping 192.168.2.2

84 bytes from 192.168.2.2 icmp_seq=1 ttl=60 time=49.858 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=60 time=51.346 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=60 time=50.083 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=60 time=48.539 ms
84 bytes from 192.168.2.2 icmp_seq=5 ttl=60 time=43.955 ms

PC5> ping 192.168.3.2

84 bytes from 192.168.3.2 icmp_seq=1 ttl=60 time=41.528 ms
84 bytes from 192.168.3.2 icmp_seq=2 ttl=60 time=45.274 ms
84 bytes from 192.168.3.2 icmp_seq=3 ttl=60 time=43.831 ms
84 bytes from 192.168.3.2 icmp_seq=4 ttl=60 time=50.020 ms
84 bytes from 192.168.3.2 icmp_seq=5 ttl=60 time=58.003 ms

PC5> ping 192.168.4.2

84 bytes from 192.168.4.2 icmp_seq=1 ttl=61 time=39.215 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=61 time=32.564 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=61 time=31.488 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=61 time=31.903 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=61 time=37.615 ms
```

```
PC4> ping 192.168.1.2
```

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=59 time=70.360 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=59 time=63.616 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=59 time=50.877 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=59 time=69.512 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=59 time=51.359 ms
```

```
PC4> ping 192.168.2.2
```

```
84 bytes from 192.168.2.2 icmp_seq=1 ttl=60 time=46.868 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=60 time=52.442 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=60 time=46.755 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=60 time=48.334 ms
84 bytes from 192.168.2.2 icmp_seq=5 ttl=60 time=59.219 ms
```

```
PC4> ping 192.168.3.2
```

```
84 bytes from 192.168.3.2 icmp_seq=1 ttl=62 time=28.526 ms
84 bytes from 192.168.3.2 icmp_seq=2 ttl=62 time=39.803 ms
84 bytes from 192.168.3.2 icmp_seq=3 ttl=62 time=20.979 ms
84 bytes from 192.168.3.2 icmp_seq=4 ttl=62 time=25.127 ms
84 bytes from 192.168.3.2 icmp_seq=5 ttl=62 time=28.789 ms
```

```
PC4> ping 192.168.5.2
```

```
84 bytes from 192.168.5.2 icmp_seq=1 ttl=61 time=41.318 ms
84 bytes from 192.168.5.2 icmp_seq=2 ttl=61 time=33.313 ms
84 bytes from 192.168.5.2 icmp_seq=3 ttl=61 time=48.990 ms
84 bytes from 192.168.5.2 icmp_seq=4 ttl=61 time=39.420 ms
84 bytes from 192.168.5.2 icmp_seq=5 ttl=61 time=39.743 ms
```

```
PC3> ping 192.168.1.2
```

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=59 time=58.990 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=59 time=68.772 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=59 time=67.358 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=59 time=61.282 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=59 time=70.568 ms
```

```
PC3> ping 192.168.2.2
```

```
84 bytes from 192.168.2.2 icmp_seq=1 ttl=60 time=59.146 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=60 time=50.293 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=60 time=49.228 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=60 time=51.655 ms
84 bytes from 192.168.2.2 icmp_seq=5 ttl=60 time=49.079 ms
```

```
PC3> ping 192.168.4.2
```

```
84 bytes from 192.168.4.2 icmp_seq=1 ttl=62 time=27.698 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=62 time=29.166 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=62 time=29.794 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=62 time=28.940 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=62 time=29.599 ms
```

```
PC3> ping 192.168.5.2
```

```
84 bytes from 192.168.5.2 icmp_seq=1 ttl=60 time=54.955 ms
84 bytes from 192.168.5.2 icmp_seq=2 ttl=60 time=58.399 ms
84 bytes from 192.168.5.2 icmp_seq=3 ttl=60 time=61.405 ms
84 bytes from 192.168.5.2 icmp_seq=4 ttl=60 time=50.002 ms
84 bytes from 192.168.5.2 icmp_seq=5 ttl=60 time=42.246 ms
```

```
PC2> ping 192.168.5.2
```

```
84 bytes from 192.168.5.2 icmp_seq=1 ttl=60 time=60.066 ms
84 bytes from 192.168.5.2 icmp_seq=2 ttl=60 time=41.212 ms
84 bytes from 192.168.5.2 icmp_seq=3 ttl=60 time=46.200 ms
84 bytes from 192.168.5.2 icmp_seq=4 ttl=60 time=52.841 ms
84 bytes from 192.168.5.2 icmp_seq=5 ttl=60 time=43.510 ms
```

```
PC2> ping 192.168.1.2
```

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=62 time=33.747 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=62 time=30.231 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=62 time=21.743 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=62 time=30.370 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=62 time=29.881 ms
```

```
PC2> ping 192.168.3.2
```

```
84 bytes from 192.168.3.2 icmp_seq=1 ttl=60 time=54.907 ms
84 bytes from 192.168.3.2 icmp_seq=2 ttl=60 time=53.443 ms
84 bytes from 192.168.3.2 icmp_seq=3 ttl=60 time=51.886 ms
84 bytes from 192.168.3.2 icmp_seq=4 ttl=60 time=60.142 ms
84 bytes from 192.168.3.2 icmp_seq=5 ttl=60 time=52.793 ms
```

```
PC2> ping 192.168.4.2
```

```
84 bytes from 192.168.4.2 icmp_seq=1 ttl=60 time=59.135 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=60 time=50.059 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=60 time=49.096 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=60 time=59.589 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=60 time=41.964 ms
```

```
PC1> ping 192.168.2.2

84 bytes from 192.168.2.2 icmp_seq=1 ttl=62 time=40.823 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=62 time=24.750 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=62 time=29.343 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=62 time=28.811 ms
84 bytes from 192.168.2.2 icmp_seq=5 ttl=62 time=30.374 ms

PC1> ping 192.168.3.2

192.168.3.2 icmp_seq=1 timeout
84 bytes from 192.168.3.2 icmp_seq=2 ttl=59 time=52.205 ms
84 bytes from 192.168.3.2 icmp_seq=3 ttl=59 time=70.106 ms
84 bytes from 192.168.3.2 icmp_seq=4 ttl=59 time=50.909 ms
84 bytes from 192.168.3.2 icmp_seq=5 ttl=59 time=52.498 ms

PC1> ping 192.168.4.2

192.168.4.2 icmp_seq=1 timeout
84 bytes from 192.168.4.2 icmp_seq=2 ttl=59 time=75.518 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=59 time=70.016 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=59 time=60.156 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=59 time=68.801 ms

PC1> ping 192.168.5.2

192.168.5.2 icmp_seq=1 timeout
84 bytes from 192.168.5.2 icmp_seq=2 ttl=59 time=58.520 ms
84 bytes from 192.168.5.2 icmp_seq=3 ttl=59 time=60.943 ms
84 bytes from 192.168.5.2 icmp_seq=4 ttl=59 time=69.136 ms
84 bytes from 192.168.5.2 icmp_seq=5 ttl=59 time=69.965 ms
```

6)

```
> Frame 9: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface -, id 0
- Ethernet II, Src: cc:04:6c:79:00:00 (cc:04:6c:79:00:00), Dst: IPv4mcast_09 (01:00:5e:00:00:09)
  > Destination: IPv4mcast_09 (01:00:5e:00:00:09)
  > Source: cc:04:6c:79:00:00 (cc:04:6c:79:00:00)
  - Type: IPv4 (0x0800)
  - [Stream index: 5]
- Internet Protocol Version 4, Src: 10.0.5.1, Dst: 224.0.0.9
  - 0100 .... = Version: 4
  - .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
  - Total Length: 52
  - Identification: 0x0000 (0)
  > 000. .... = Flags: 0x0
  - ...0 0000 0000 0000 = Fragment Offset: 0
  > Time to Live: 2
  - Protocol: UDP (17)
  - Header Checksum: 0xc8ef [validation disabled]
  - [Header checksum status: Unverified]
  - Source Address: 10.0.5.1
  - Destination Address: 224.0.0.9
  - [Stream index: 2]
- User Datagram Protocol, Src Port: 520, Dst Port: 520
  - Source Port: 520
  - Destination Port: 520
  - Length: 32
  - Checksum: 0x49e5 [unverified]
  - [Checksum Status: Unverified]
  - [Stream index: 1]
  - [Stream Packet Number: 1]
  > [Timestamps]
  - UDP payload (24 bytes)
- Routing Information Protocol
  - Command: Response (2)
  - Version: RIPv2 (2)
  > IP Address: 192.168.1.0, Metric: 1
```



```

> Frame 2: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
  Ethernet II, Src: cc:03:6c:5c:00:00 (cc:03:6c:5c:00:00), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
    > Destination: IPv4mcast_05 (01:00:5e:00:00:05)
    > Source: cc:03:6c:5c:00:00 (cc:03:6c:5c:00:00)
    Type: IPv4 (0x0800)
    [Stream index: 1]
  Internet Protocol Version 4, Src: 10.0.0.3, Dst: 224.0.0.5
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    > Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
    Total Length: 84
    Identification: 0x1ab1 (6833)
    > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 1
    Protocol: OSPFIGP (89)
    Header Checksum: 0xb3d8 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.0.0.3
    Destination Address: 224.0.0.5
    [Stream index: 0]
  Open Shortest Path First
    > OSPF Header
    > OSPF Hello Packet
    > OSPF LLS Data Block

```

7)

```

> ls
rt_R1.cfg rt_R2.cfg rt_R3.cfg rt_R4.cfg rt_R5.cfg rt_R6.cfg rt_R7.cfg rt_R8.cfg
~/Projects/eltex_summer_course/module4/lab5 > on main !1 ?2

```

8)

```

> ls
R1.cfg R3.cfg R5.cfg R7.cfg rt_R1.cfg rt_R3.cfg rt_R5.cfg rt_R7.cfg
R2.cfg R4.cfg R6.cfg R8.cfg rt_R2.cfg rt_R4.cfg rt_R6.cfg rt_R8.cfg
~/Projects/eltex_summer_course/module4/lab5 > on main !1 ?2

```